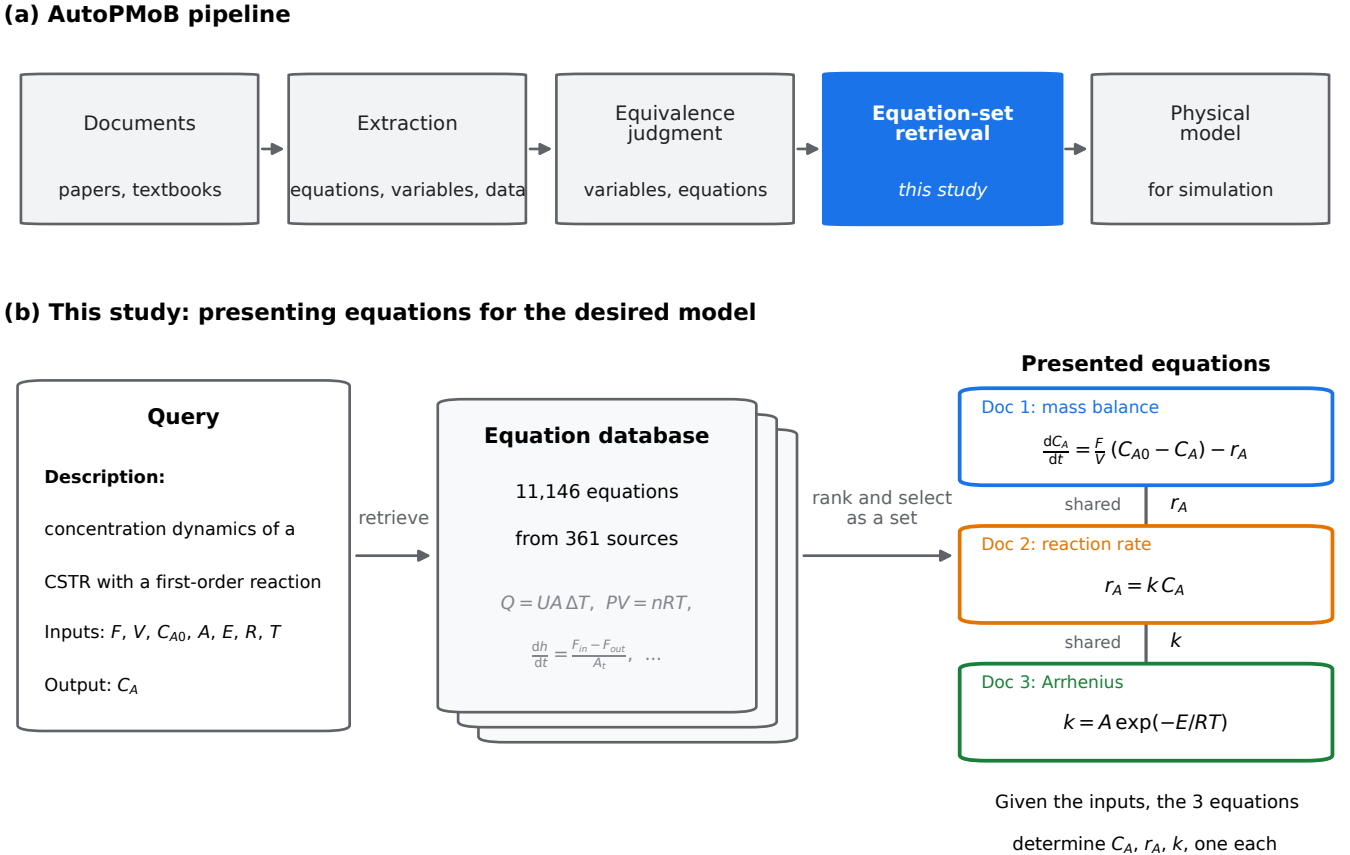
(a) AutoPMoB pipeline
Documents
papers, textbooks
Extraction
equations, variables, data
Equivalence judgment
variables, equations
Equation-set retrieval
this study
Physical model
for simulation
(b) This study: presenting equations for the desired model
Query
Description:
concentration dynamics of a CSTR with a first-order reaction
Inputs: F, V, C_A0, A, E, R, T
Output: C_A
retrieve
Equation database
11,146 equations
from 361 sources
Q = UA ΔT, PV = nRT,
dh/dt = (F_in − F_out)/A_t, ...
rank and select as a set
Presented equations
Doc 1: mass balance
dC_A/dt = F/V (C_A0 − C_A) − r_A
shared
r_A
Doc 2: reaction rate
r_A = k C_A
shared
k
Doc 3: Arrhenius
k = A exp(−E/RT)
Given the inputs, the 3 equations
determine C_A, r_A, k, one each